

The proposed model performs multimodal myocardial infarction classification by jointly leveraging ECG time-series, ECG images, and laboratory biomarkers. A Graphormer encoder extracts inter-lead temporal features from ECG signals, while a Vision Transformer captures morphological patterns from ECG images. Both branches are dynamically modulated by laboratory features via FiLM layers to enable patient-specific encoding. The representations are then integrated through a Transformer-based fusion module that captures fine-grained cross-modal interactions. A contrastive learning objective further aligns the modalities in a shared latent space, enhancing robustness and diagnostic accuracy.